PROTEIN PARADOX:

Health, Habits, and Hype



A project by: Rasulov Abdulvoris

QUESTIONS

Are we getting healthier, or just more obsessed with protein?

- Does a higher daily protein supply mean a lower obesity rate?
- How much does physical inactivity impact global overweight percentages?
- Is the world actually becoming more interested in "Protein"?
- What kind of protein are they eating
- Which countries care the most about it?



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THE DATASET

AND CLEANING

kaggle + Our World in Data + = master_dataset

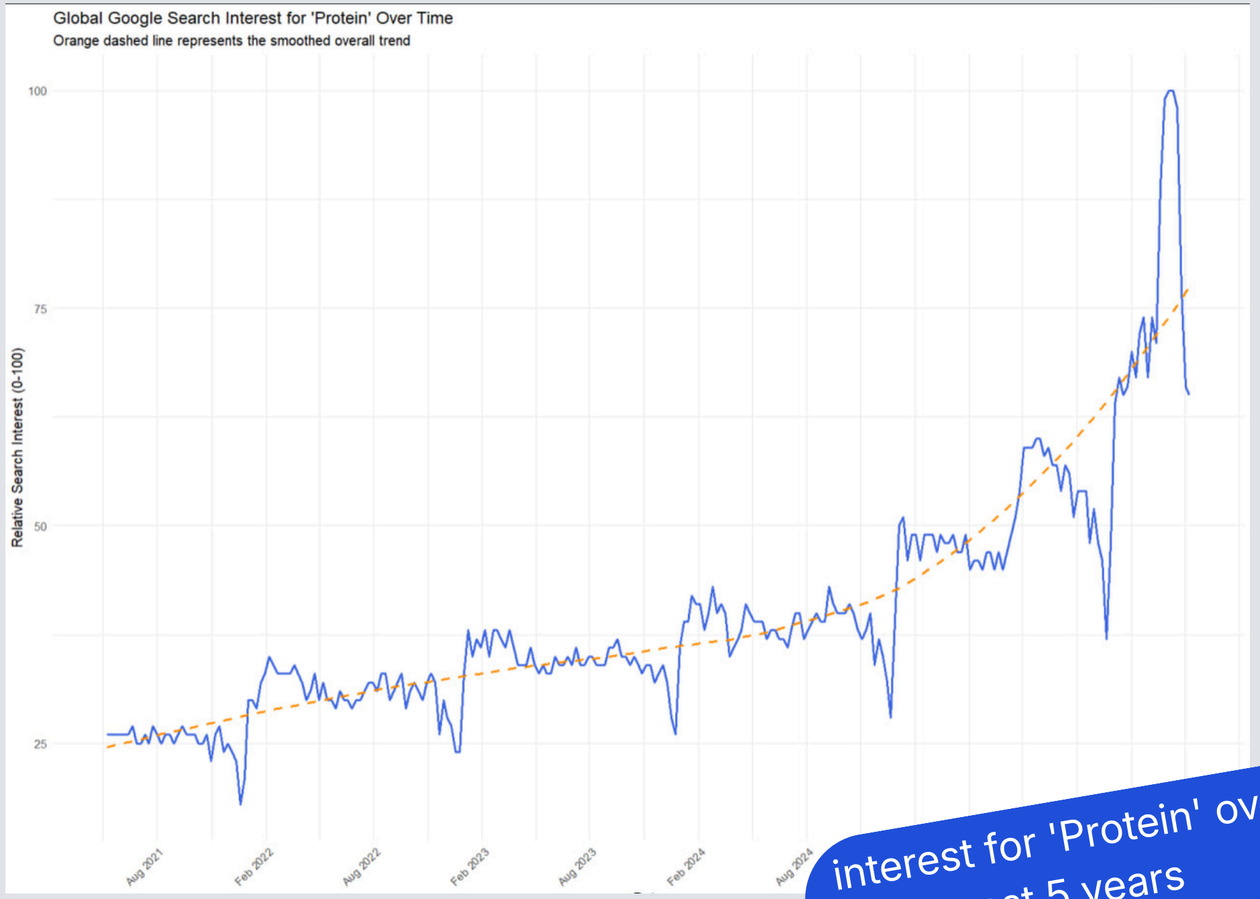
Afghanistan	Eastern Mediterranean	2022	33.36	FALSE	2022	48.57364	FALSE	2023	0.1200000
Angola	Africa	2022	17.33	FALSE	2022	31.77765	FALSE	2023	4.4800000
Albania	Europe	2022	24.28	FALSE	2022	62.16242	FALSE	2023	2.5100000
United Arab Emirates	Eastern Mediterranean	2022	66.14	TRUE	2022	71.32877	FALSE	2023	6.1499950
Argentina	Americas	2022	38.51	FALSE	2022	68.42044	FALSE	2023	1.9399999
Armenia	Europe	2022	26.58	FALSE	2022	55.42755	FALSE	2023	1.4799960
Antigua and Barbuda	Americas	2022	38.12	FALSE	2022	62.66753	FALSE	2023	13.5284050
Australia	Western Pacific	2022	25.47	FALSE	2022	63.95607	FALSE	2023	6.1099987
Austria	Europe	2022	19.76	FALSE	2022	44.98951	FALSE	2023	3.6499984
Azerbaijan	Europe	2022	23.92	FALSE	2022	61.68490	FALSE	2023	0.6200001
Burundi	Africa	2022	14.27	FALSE	2022	16.31177	FALSE	2022	0.7353806
Belgium	Europe	2022	25.41	FALSE	2022	51.07649	FALSE	2023	5.5999994
Benin	Africa	2022	18.02	FALSE	2022	30.07517	FALSE	2022	4.3281900
Burkina Faso	Africa	2022	16.94	FALSE	2022	21.70263	FALSE	2023	2.9700003
Bangladesh	South-East Asia	2022	20.30	FALSE	2022	27.02097	FALSE	2023	7.9000000
Bulgaria	Europe	2022	32.28	FALSE	2022	51.94807	FALSE	2023	2.1699984
Bahrain	Eastern Mediterranean	2022	38.61	FALSE	2022	68.68815	FALSE	2023	5.1299872
Bahamas	Americas	2022	37.17	FALSE	2022	75.78784	FALSE	2023	7.5697727
Bosnia and Herzegovina	Europe	2022	21.07	FALSE	2022	56.85756	FALSE	2023	1.7399989
Belarus	Europe	2022	14.02	FALSE	2022	55.41775	FALSE	2023	3.3300002
Belize	Americas	2022	41.52	FALSE	2022	72.77852	FALSE	2023	3.5799823
Bolivia	Americas	2022	22.57	FALSE	2022	65.14485	FALSE	2023	0.6399999
Brazil	Americas	2022	40.40	FALSE	2022	62.99998	FALSE	2023	2.2600000
Barbados	Americas	2022	44.03	FALSE	2022	67.16144	FALSE	2023	11.8597060
Bhutan	South-East Asia	2022	9.91	FALSE	2022	46.67578	FALSE	2023	1.8799857
Botswana	Africa	2022	22.28	FALSE	2022	41.99094	FALSE	2023	0.5699995
Canada	Americas	2022	37.18	FALSE	2022	59.15438	FALSE	2023	4.9700003

- **Ordered the data and dropped rows N/A entries**
- **Remove or add columns and rows**
- **merge into a single "Master Dataframe" matched by country names and regions**

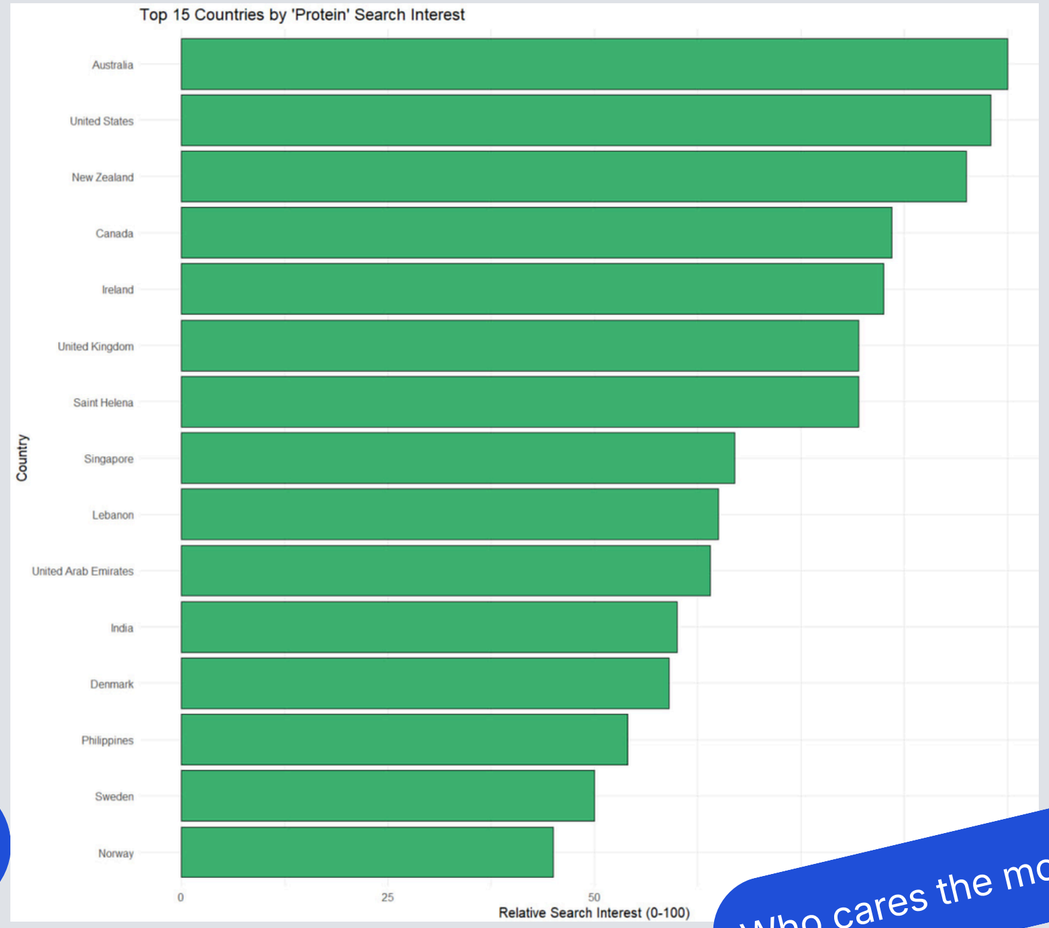
Analysis reflects the most recent data available per country up to 2023

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THE GLOBAL HYPE



interest for 'Protein' over past 5 years

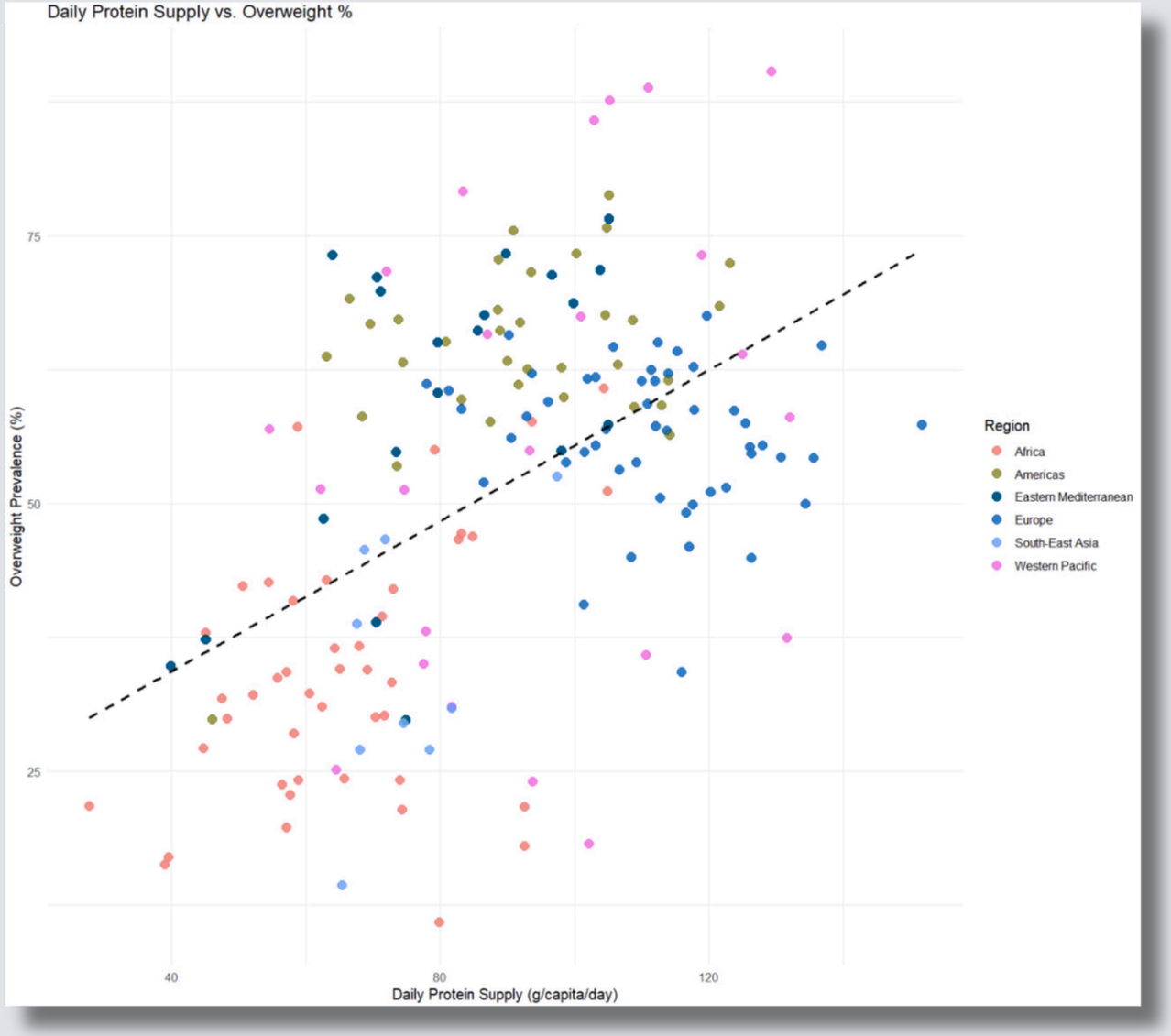


Who cares the most??

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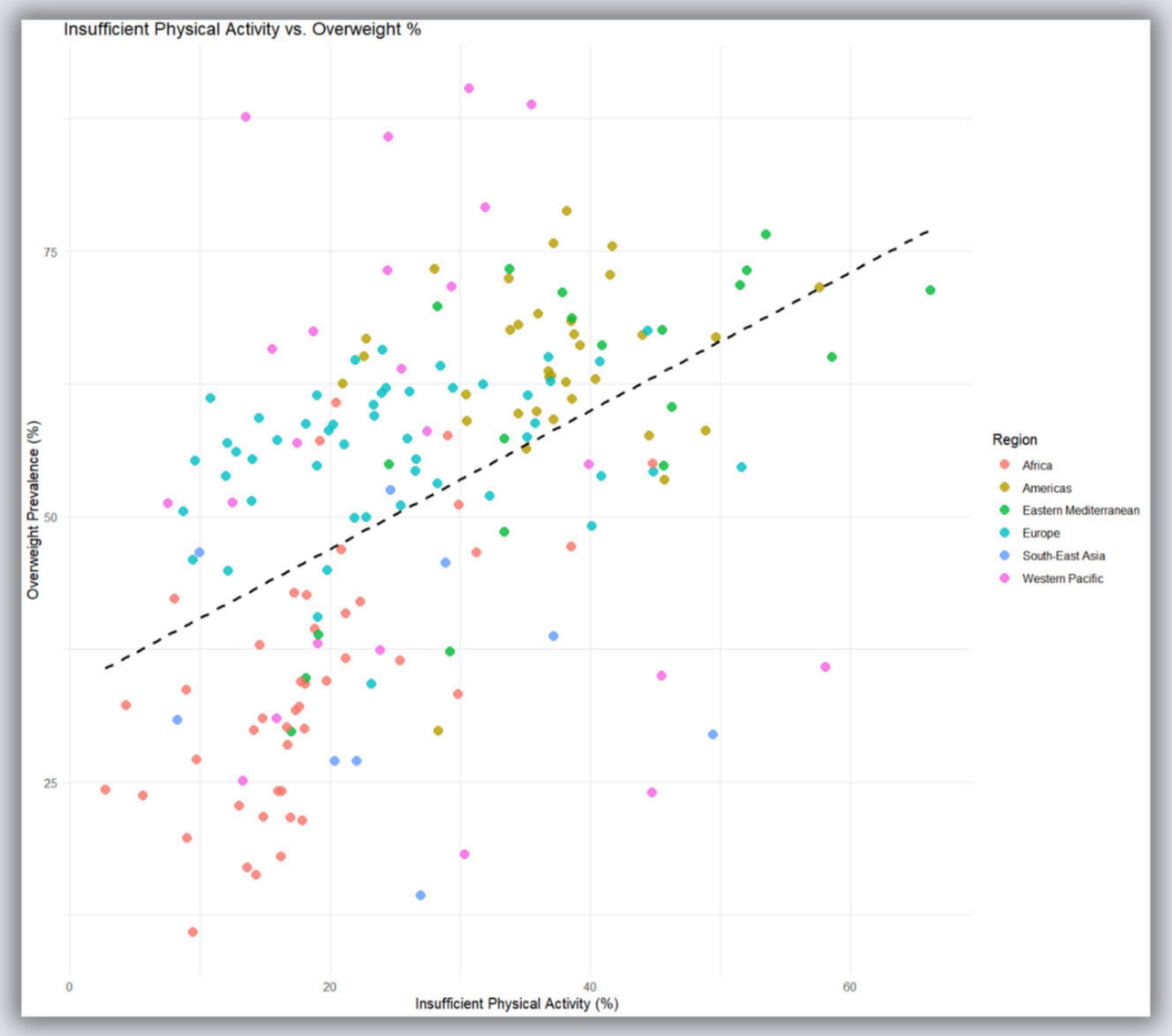
PROTEIN SUPPLY & OVERWEIGHT ↗

This plot shows a strong positive correlation telling us that, on a global scale countries with a higher daily supply of protein per person tend to have a higher percentage of their population classified as overweight



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INSUFFICIENT PHYSICAL ACTIVITY & OVERWEIGHT



Here, we also see a positive correlation, though it looks slightly more spread out. As the percentage of people getting insufficient physical activity goes up, the percentage of overweight people also tends to go up.

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TEST STATISTIC

```
Call:
lm(formula = Overweight_Pct ~ Daily.protein.supply + Physical_Activity_Insufficient_Pct,
    data = master_df)

Residuals:
    Min      1Q  Median      3Q     Max
-38.800  -7.215   0.674   8.024  38.480

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                      11.89207    3.96727   2.998  0.00311 **
Daily.protein.supply              0.28833    0.04174   6.907 8.56e-11 ***
Physical_Activity_Insufficient_Pct  0.51617    0.08113   6.362 1.65e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.28 on 177 degrees of freedom
Multiple R-squared:  0.3961,    Adjusted R-squared:  0.3893
F-statistic: 58.05 on 2 and 177 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression

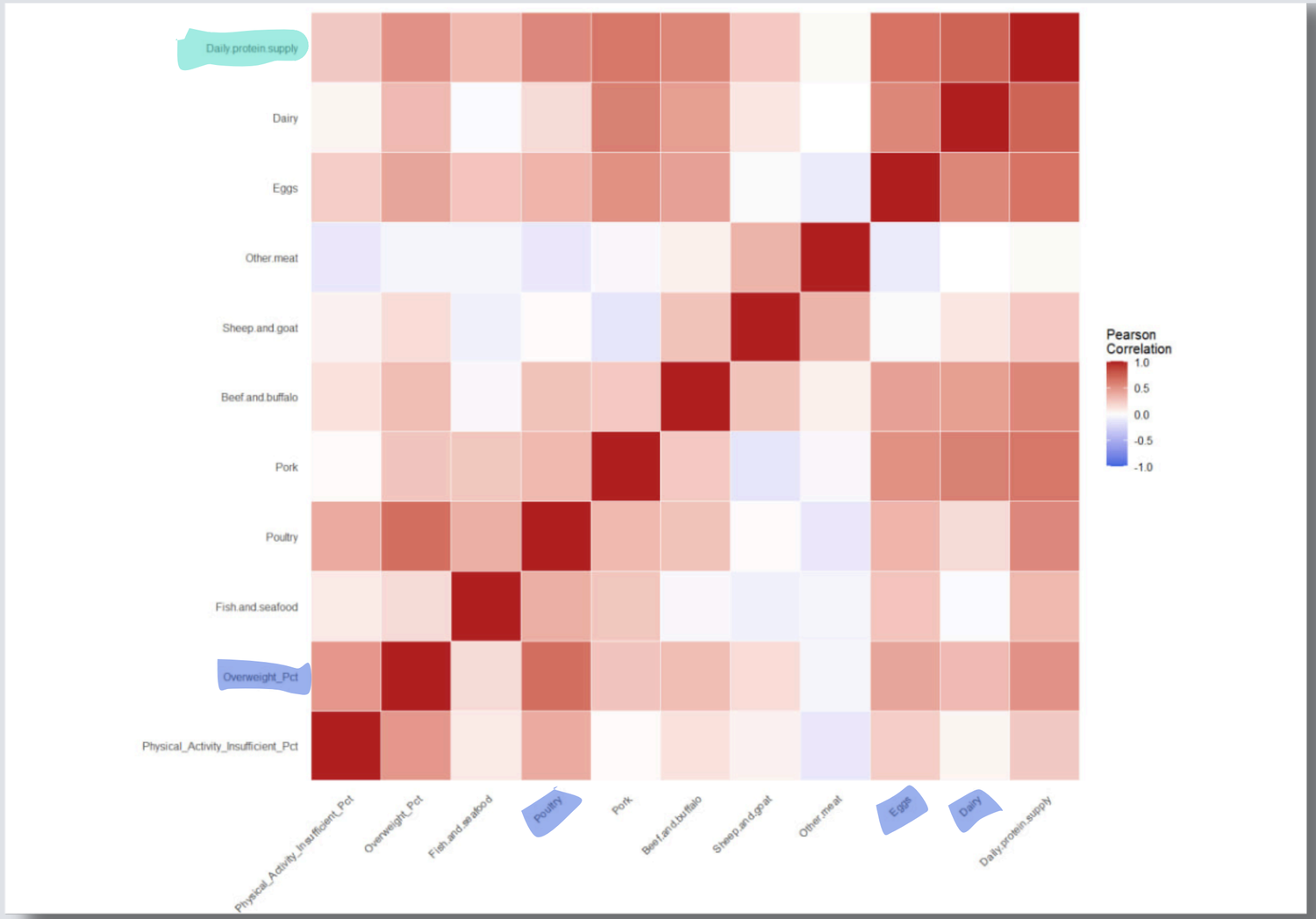
1. **Slope** explains how much the overweight percentage changes when we increase a predictor by one unit, assuming the other variable stays exactly the same

2. The **R-squared** is ~0.396: This means that protein intake and physical activity together explain roughly 40% of the variance in a country's overweight prevalence.

3. **P-value** results are highly statistically significant. The probability that these relationships are just a random coincidence is essentially zero

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COORELATION HEATMAP



1 This heatmap reveals that almost everything has a positive relationship with obesity, but specifically Poultry, Eggs, and Dairy have noticeably darker red squares compared to others

2 When a country has a very high daily protein supply, this heatmap shows that the surplus is heavily driven by dairy and egg consumption, rather than just meat

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CONCLUSION

People are searching for "protein" more than ever to get healthy. However, data showed that countries eating the most protein actually have the highest obesity rates. This may mean that high protein intake usually just means eating a lot more calories overall.



For future analyses it would be interesting to check how economics play a role specifically, whether the cost for proteins directly influences a country's obesity percentage.

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THANK YOU

